
Supplementary information

Giant enhancement of third-harmonic generation in graphene–metal heterostructures

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Giant enhancement of third-harmonic generation in graphene-metal heterostructures

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SUPPLEMENTARY INFORMATION

A. Supplementary Figures

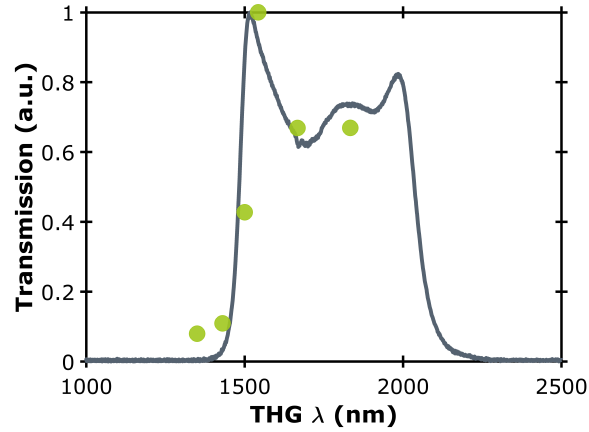


Figure 1. **Confirmation of the third-harmonic signal wavelength with a 5500 nm incident beam.** The transmission of the third-harmonic signal through a bandpass filter at 1750 nm with a 500 nm bandwidth as the incident wavelength is changed. The points are the measured transmission of the third-harmonic signal and the solid curve is the transmission of the filter measured with standard FTIR.

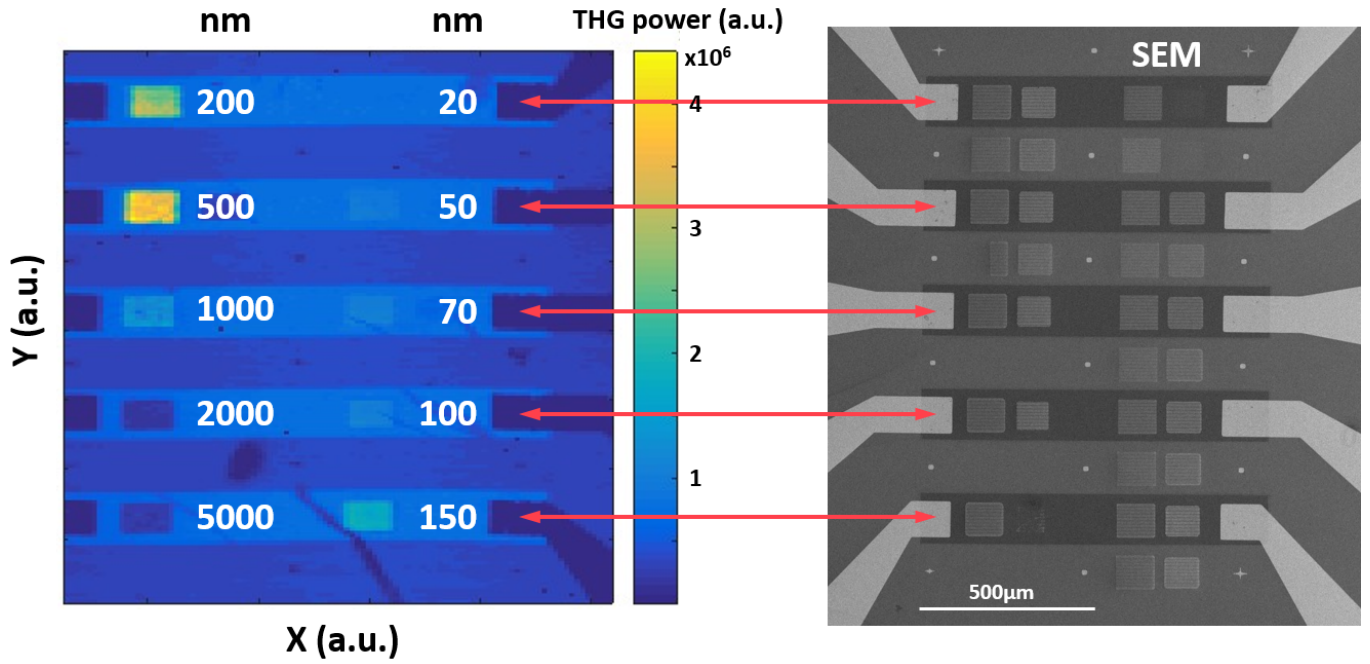


Figure 2. **Comparison of the third-harmonic signal with an SEM image of the sample.** The left panel shows the third-harmonic signal as the entire sample is moved transversely through the incident beam. The corresponding SEM image of the sample is shown on the right. Strong third-harmonic signals can be distinguished in the regions with both graphene and golden nanoribbons, as well as a weaker signals from the strips of bare graphene surrounding the nanoribbons. There is no discernible signal from the nanoribbons without graphene. The darker areas indicating the contacts are also evident. As a reference, the SEM image on the right hand side confirms the structure of the sample, where the dark horizontal bands indicate the presence of bare graphene. The brighter squares on top of these bands are different arrays of golden nanoribbons with widths W ranging from 20 – 5000 nm. Note that some of the arrays noticeable on the SEM image do not show any THG signal. This is due to experimental imperfections of the golden nanoribbons during the fabrication process.

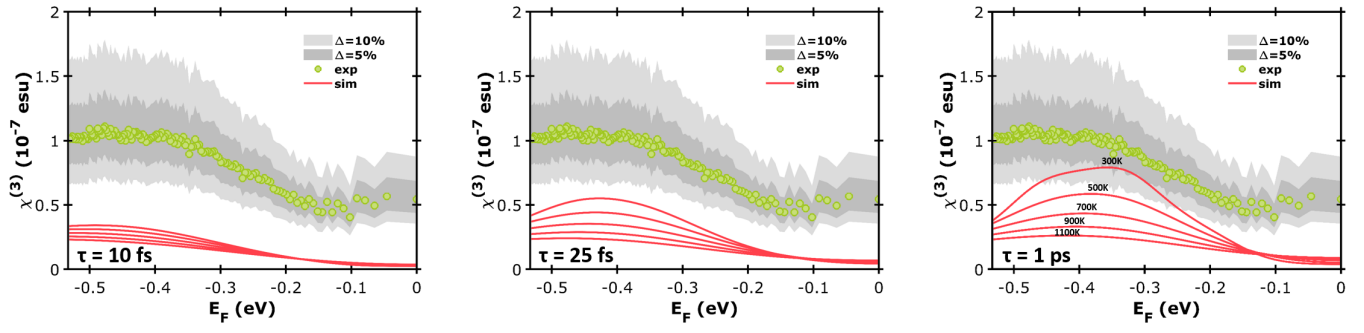


Figure 3. Comparison of the magnitude of the experimentally estimated $\chi^{(3)}$ with the calculated value. The shaded areas around the green points represent the experimental uncertainty in our estimation of $\chi^{(3)}$. We calculate the error bars of the experimental data (green dots) with an uncertainty of 5% and 10% on the experimentally measured parameters, defined by the dark and light grey areas, respectively. The red solid lines show the theoretically calculated $\chi^{(3)}$ at different electron temperatures for 10 fs, 25 fs and 1 ps, from left to right.

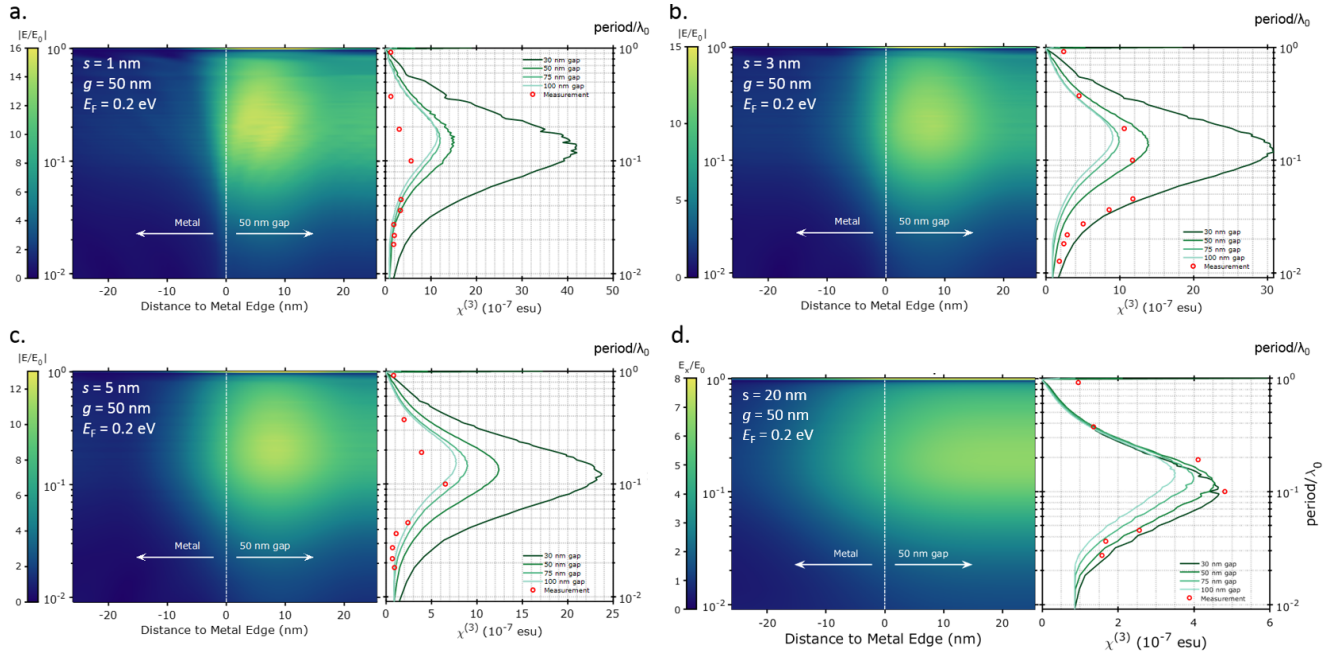


Figure 4. **Field confinement in the gap between the nanoribbons and $\chi^{(3)}$ as a function of the nanoribbon width.** Each panel corresponds to a different Al_2O_3 spacing $s = [1, 3, 5, 20]$ nm.

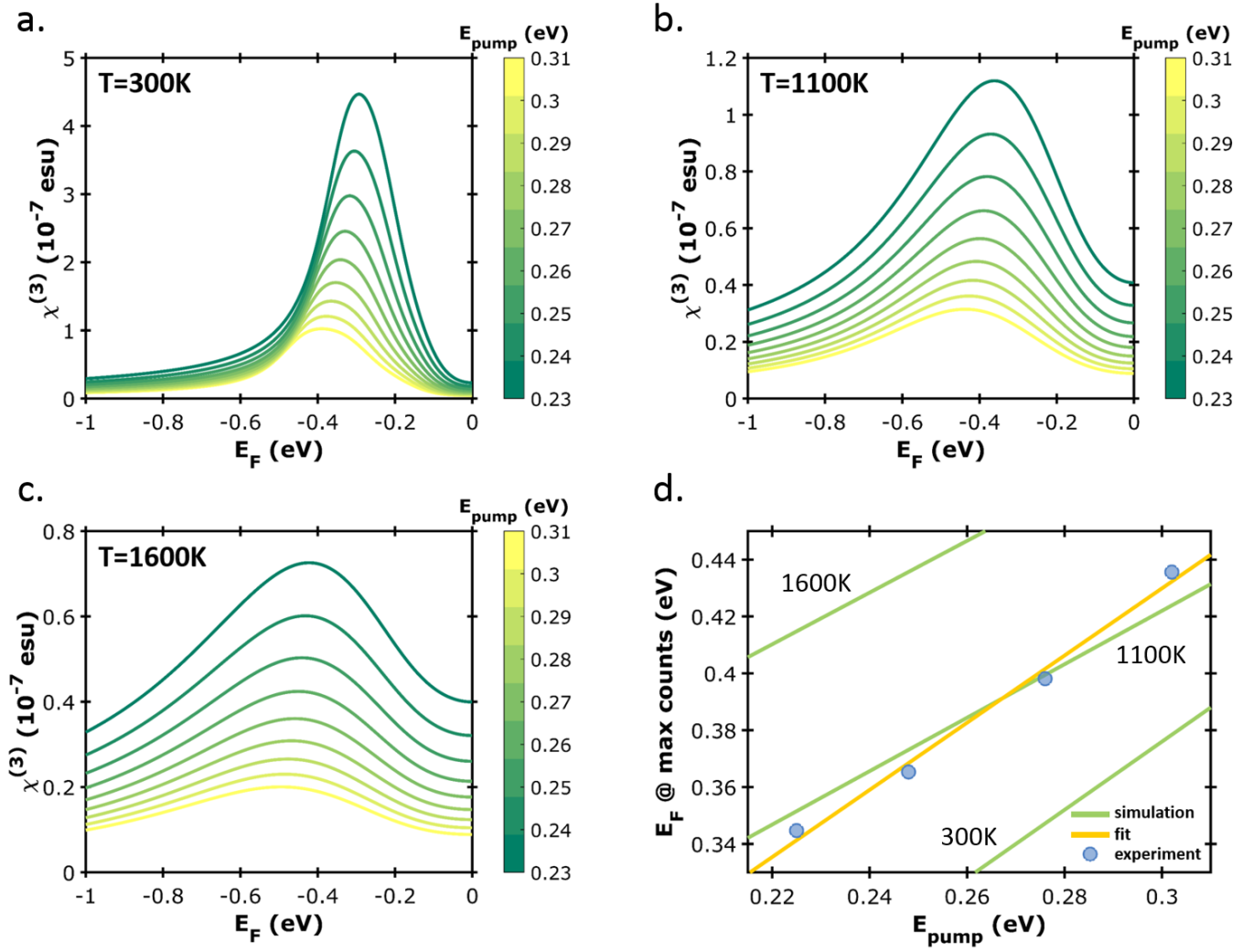


Figure 5. **Effect of the electron temperature on the optimum Fermi energy.** **a-c.** Simulations of the third-order susceptibility $\chi^{(3)}$ as a function of the Fermi energy E_F for incident wavelengths in the $\lambda_0 = 4.1 - 5.5 \mu\text{m}$ ($E_0 = 0.225 - 0.302$ eV) range and electron temperatures of 300 K, 1100 K and 1600 K, respectively. **d.** Blue dots show the experimental values of the E_F at which the maximum $\chi^{(3)}$ are found in the gate measurement, for $\lambda_0 = [5.5, 5.0, 4.5, 4.1] \mu\text{m}$ ($E_0 = [0.225, 0.248, 0.276, 0.302]$ eV). The green solid lines are simulations at $T_e = [300, 1100, 1600]$ K and the solid yellow line is a linear fit of the data points.

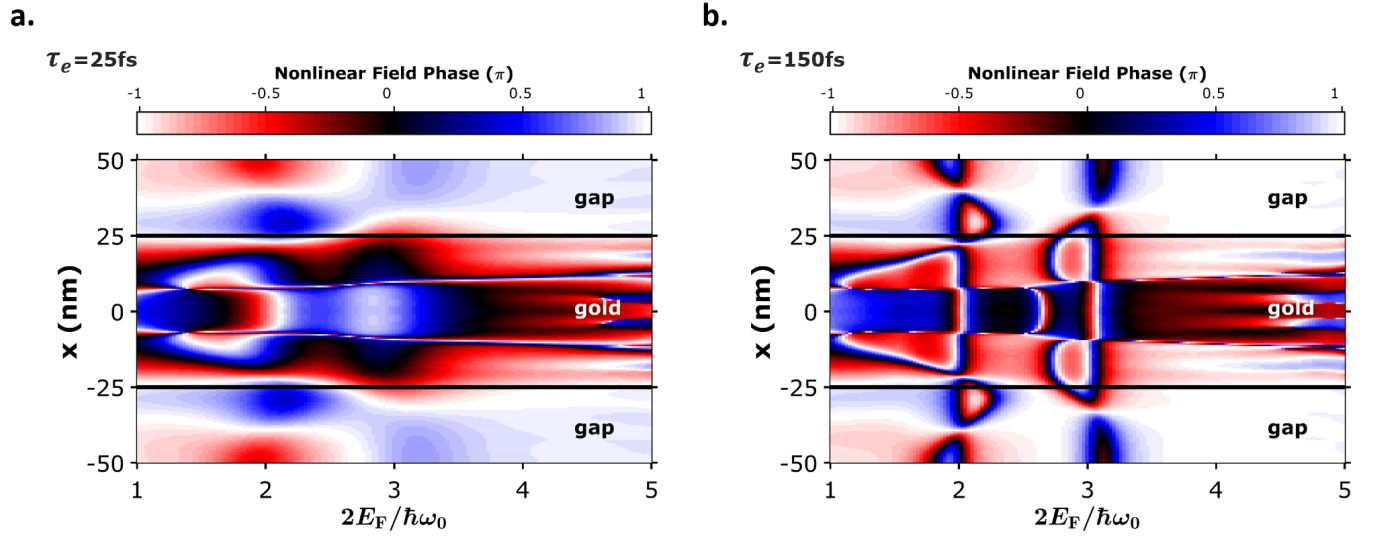


Figure 6. **Phase of the electric field in graphene.** Phase of the simulated $|E_x(\omega)^3 E_x(3\omega)/E_0|$ field in the graphene layer below the gold nanoribbons and in the gap, as a function of E_F periods for $\lambda_0 = 5.5\mu\text{m}$, and plasmon lifetime of **a.** $\tau_e = 25\text{fs}$, **b.** $\tau_e = 150\text{fs}$.

B. Rigorous coupled-wave analysis (RCWA)

The simulations in Fig. 3a and 4 of the main text are achieved with the Matlab script from Ref.^{1,2} The geometry of the 2D simulation is given by the layer stack of the original chip, which is composed by a $525\text{ }\mu\text{m}$ -thick silicon chip coated with a 285 nm -thick SiO_2 layer above and below. The monolayer graphene placed on the front side of the chip is modelled by a surface nonlocal conductivity, given in Ref.³ The spacer layer built by h-BN or Al_2O_3 with thickness s was placed on top of the graphene. Finally a 10 nm -thick gold nanoribbon with width W was placed on top of the spacer. In a good approximation, the electric permittivity for silicon was assumed to be 12 and wavelength independent. The wavelengths depending dielectric permittivity of SiO_2 , h-BN and gold are found in Ref.,⁴ Ref.⁵ and Ref.,⁶ respectively. As mentioned in the Methods, the simulation script requires a periodic structure in the lateral dimension. Here the periodicity is given by the sum of the nanoribbon width and the gap in-between. The electric field of the excitation incident light (THG signal) in the graphene layer was calculated for a normal incident of a monochromatic plane wave polarized parallel to the x axis and angular frequency ω (3ω). From the electric fields $E_x(\omega)$ and $E_x(3\omega)$, $|\chi_{\text{sim}}^{(3)}|$ was calculated as described in the Methods section.

For accessing the electron temperature T_e in the graphene layer, we calculate the amount of absorbed energy in the graphene. To do so, we run the simulations with the RCWA, considering an ambient temperature of 300 K (see Fig. 7a-c. Solving the implicit Eq. 8 – 10 of the main text leads to the electron temperature used in Fig. 4. These temperatures were considered in the third-order nonlinear conductivity and are shown in Fig. 7d-f. During the gating measurements, when the voltage is applied between the graphene and silicon substrate, a leakage current through the SiO_2 substrate can occur. Because of that, the charge densities in the graphene layer may differ slightly from those determined via the capacitor model introduced in the Methods section. This difference depends on the measurement time and the leakage. In order to compensate for this effect, the gating voltage of the experimental data shown in Fig. 4a-c of the main text has been fitted accordingly.

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- ¹ <https://es.mathworks.com/matlabcentral/fileexchange/55401-ppml-periodically-patterned-multi-layer> (31.10.2019).
- ² Manceau, J.-M., Zanutto, S., Sagnes, I., Beaudoin, G. & Colombelli, R. Optical critical coupling into highly confining metal-insulator-metal resonators. *Applied Physics Letters* **103**, 091110 (2013).
- ³ Koppens, F. H., Chang, D. E. & García de Abajo, F. J. Graphene plasmonics: A platform for strong light-matter interactions. *Nano Lett.* **11**, 3370–3377 (2011).
- ⁴ Kischkat, J. *et al.* Mid-infrared optical properties of thin films of aluminum oxide, titanium dioxide, silicon dioxide, aluminum nitride, and silicon nitride. *Applied optics* **51**, 6789–6798 (2012).
- ⁵ Caldwell, J. D. *et al.* Sub-diffractive volume-confined polaritons in the natural hyperbolic material hexagonal boron nitride. *Nature communications* **5**, 1–9 (2014).
- ⁶ Olmon, R. L. *et al.* Optical dielectric function of gold. *Physical Review B* **86**, 235147 (2012).

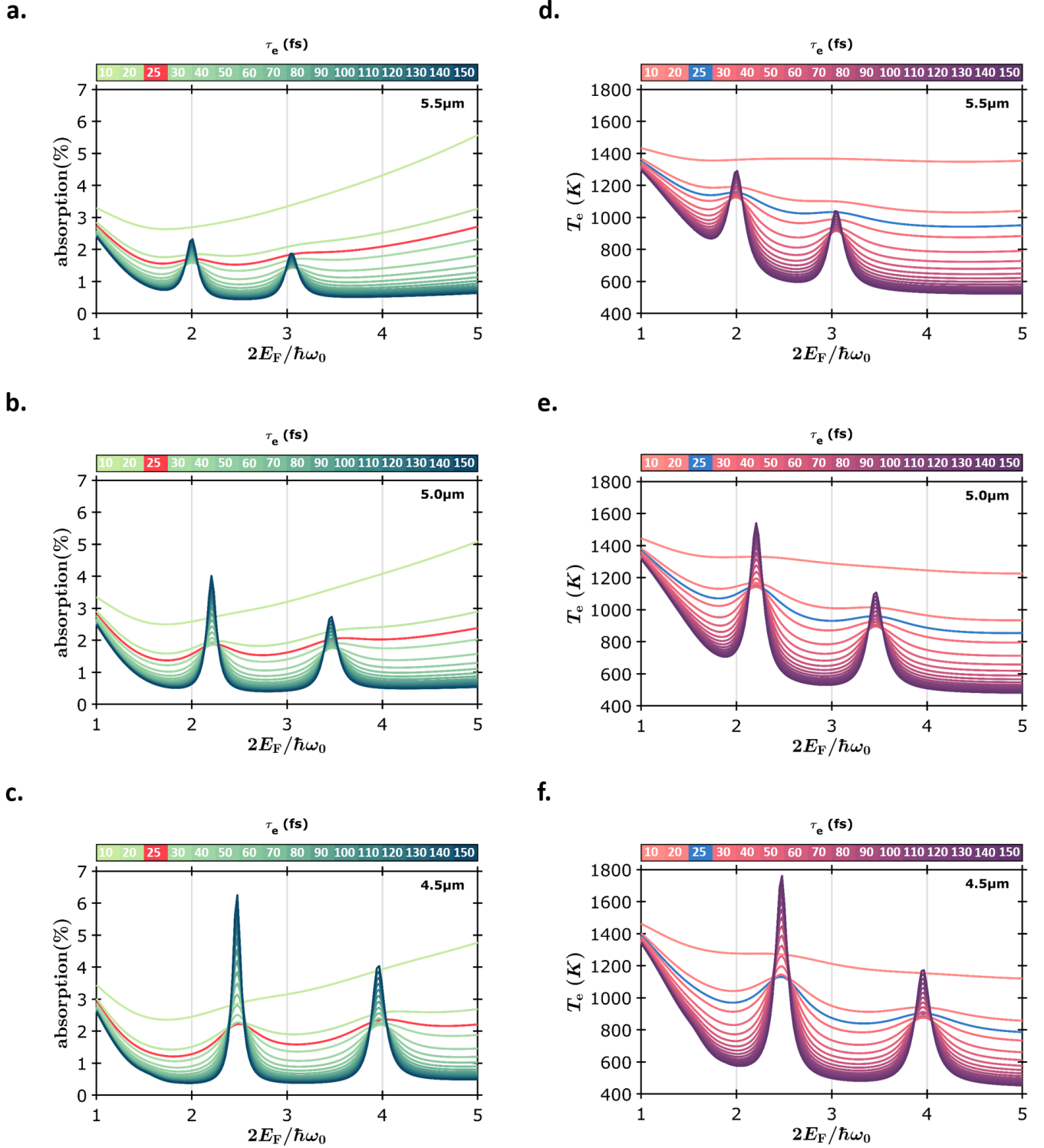


Figure 7. **Light absorption and electron temperature.** a.-c. Absorption of the excitation field in graphene for incident wavelengths 5.5 μm , 5.0 μm and 4.5 μm , respectively. d.-f. Electron temperature T_e derived from the absorbed excitation field in graphene for incident wavelengths 5.5 μm , 5.0 μm and 4.5 μm , respectively.